Chain Necklace and "Controlled Threat": A Pilot Similarity Test in CLIP (ViT-B/32)

Ege Şırlakoğlu
University of London

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Abstract

This report tests similarity with a vision-language model. Study uses CLIP (ViT-B/32) and cosine similarity in the embedding space (Radford et al., 2021). I am testing one visual cue and it is a chain necklace. I am asking if images of men wearing a chain necklace are closer to guard dog concepts than similar male images without a chain necklace. I also test a second group about boundary respect and responsiveness. Stimuli and licensing details are included in the `stimuli.csv` file and image-text similarities are in `similarity_matrix.csv`. Group means and simple deltas are reported. In this stimulus set, chain-necklace images were not closer to the guard group than neutral male images. They were also not closer to the boundary group. Image-image check also showed the same direction. This is a small pilot study, so results are descriptive.

Keywords: CLIP; cosine similarity; vision–language model; representational similarity

1 Introduction

Vision-language models place images and texts in one shared space. In model space, it is a black box, so understanding without testing is not possible. Vision-language models are not humans and they do not predict human behavior and thinking. This can be helpful for people who do not have real human data, but they have computer data.

In this pilot study, I am testing one specific cue and it is a chain necklace. Comparing images of adult men wearing a chain necklace to images of adult men without a chain necklace, I measured similarity to two short families. The first family describes a guard or aggressive dog and the second family describes a protective person who respects boundaries and responds to feedback.

2 Method

2.1 Stimuli

In this study, I used 12 open-licensed images listed in the `stimuli.csv` file. It includes the image URLs, licenses and attribution text. It also includes brief notes on possible confounds. Stimuli were grouped into four categories (three images each): `man_chain`, `man_neutral`, `dog_chain_guard`, and `dog_neutral`. In this report, the main comparison is between `man_chain` and `man_neutral`.

2.2 Text prompts

I used eight short prompts. They cover (a) guard or aggressive dog language, (b) boundary-respecting and responsive language, and (c) neutral controls. The full prompt list is in Appendix A.1.

2.3 Model and similarity

I used CLIP with the ViT-B/32 image encoder and its matching text encoder (Radford et al., 2021). For each image, I extracted an image embedding. For each prompt, I extracted a text embedding. I then computed cosine similarity for each image-text pair. The full matrix is saved in `similarity_matrix.csv`.

2.4 Group summaries

I report averages within stimulus groups and within prompt families. I also report simple differences (deltas) between `man_chain` and `man_neutral`. I report these values exactly as computed in the project outputs.

3 Results

Figure 1 shows the full image-text similarity pattern as a heatmap. Figure 2 shows the main contrasts as a summary bar plot. Table 1 reports the key group means and deltas. I also ran an image-image similarity check. The computed means were:

`avg_man_chain_guard = 0.1688338`, `avg_man_chain_boundary = 0.2160057`, `avg_man_neutral_guard = 0.1832857`, and `avg_man_neutral_boundary = 0.2194173`. For the guard prompt family, `man_chain` had an average similarity of 0.1688338, while `man_neutral` had 0.1832857. The delta was $\Delta_{\text{guard}} = -0.0144519$. For the boundary prompt family, `man_chain` had 0.2160057 and `man_neutral` had 0.2194173. The delta was $\Delta_{\text{boundary}} = -0.0034116$.

We also ran an image–image similarity check. The average similarity from `man_chain` images to `dog_guard` images was 0.4443055. For `man_neutral` to `dog_guard`, it was 0.5546486. The delta was $\Delta_{\text{imgimg}} = -0.1103431$.

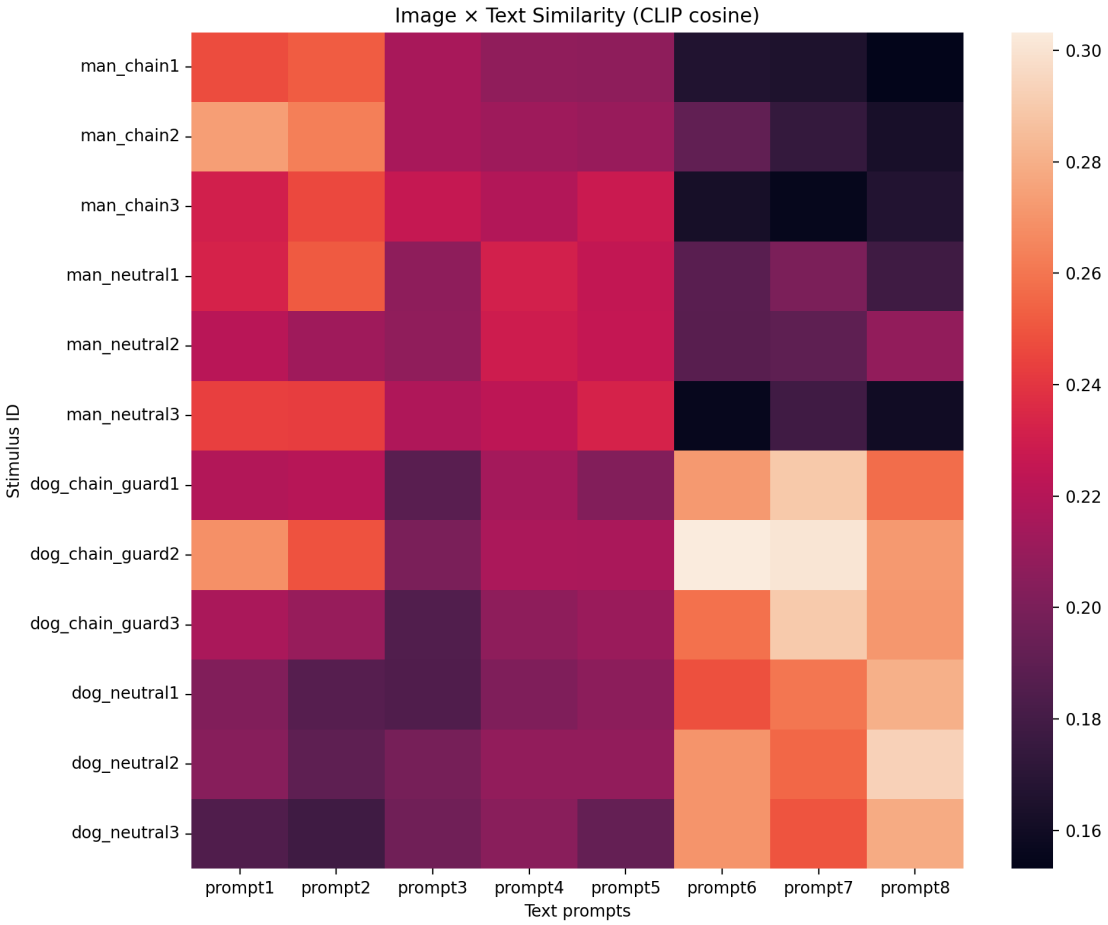

Figure 1. Image × text cosine similarity heatmap (CLIP ViT-B/32).

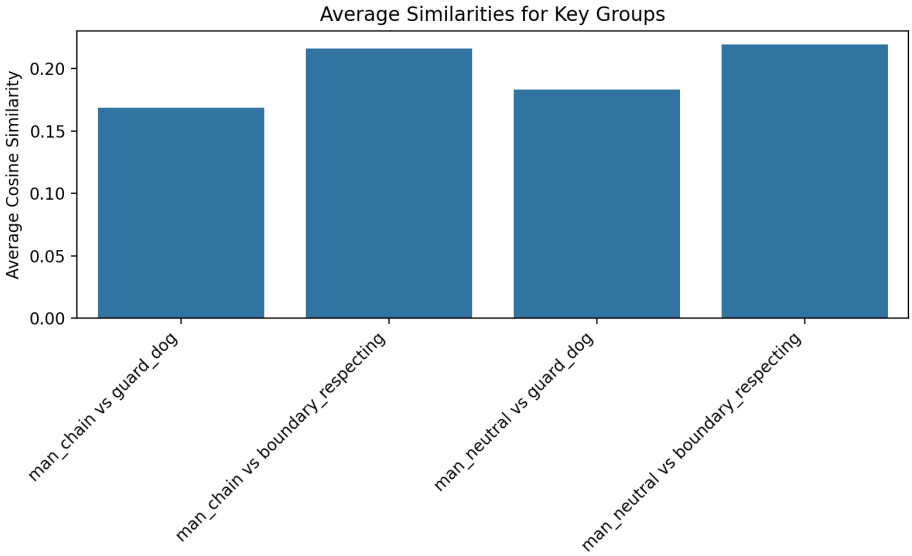

Figure 2. Average cosine similarities for the key group contrasts.

Table 1. Summary similarities and deltas.

Comparison	Man chain	Man neutral	Δ (chain minus neutral)
Guard prompt family (image–text)	0.1688338	0.1832857	-0.0144519
Boundary prompt family (image–text)	0.2160057	0.2194173	-0.0034116
Dog guard concept (image–image)	0.4443055	0.5546486	-0.1103431

4 Discussion

This pilot did not support the hypothesis for this stimulus set. Results show that chain necklace images were not closer to the guard dog group than the neutral male images. They were also not closer to the boundary prompt group. Image-image check also showed the same direction.

There might be a couple of reasons. For instance, the stimulus set is very small, it has three images per category. It was also hard to match confounds in a small set such as age, lighting, clothing style, and photo style. It might also be hard to infer consent, boundaries, and responsiveness from a still image.

5 Open Materials and Availability

The project includes `stimuli.csv`, `similarity_matrix.csv`, `similarity_heatmap.png`, `summary_barplot.png`, and `analysis.ipynb`. Images are referenced by URL with license and attribution in `stimuli.csv`.

A Appendix

A.1 Text prompts

We used the following eight prompts:

1. A man wearing a chain necklace showing a dominant and protective demeanor, ready to defend his partner.
2. A man wearing a chain necklace who appears dangerous and uncontrollable, projecting aggression.
3. A dominant man who still respects boundaries and values consent.
4. A protective man who is responsive and attentive to his partner's needs.
5. A gentle or submissive man showing controllable behavior and deference.
6. A loyal guard dog on a chain, alert and ready to protect.
7. An aggressive dog restrained by a chain, barking and showing threat.
8. A calm dog without a chain, relaxed and friendly.

A.2 Stimulus attribution list (from `stimuli.csv`)

References

Radford, A., Kim, J. W., Hallacy, C., Ramesh, A., Goh, G., Agarwal, S., Sastry, G., Askell, A., Mishkin, P., Clark, J., Krueger, G., & Sutskever, I. (2021). Learning transferable visual models from natural language supervision (arXiv:2103.00020). arXiv. https://arxiv.org/abs/2103.00020

Table 2. Stimuli, licenses, and attributions.

ID	License	Attribution text	URL
man_chain1	CC BY-SA 3.0	Photo by Newsact11 via Wikimedia Commons (CC BY-SA 3.0)	https://upload.wikimedia.org/wikipedia/commons/3/3f/Actor_Kevin_Marc.jpg
man_chain2	CC BY-SA 4.0	Photo by Joseph Ades via Wikimedia Commons (CC BY-SA 4.0)	https://upload.wikimedia.org/wikipedia/commons/thumb/4/4e/Casanova_(August_2016).jpg/1280px-Casanova_(August_2016).jpg
man_chain3	CC BY-SA 4.0	Photo by Ernest Ankomah Frimpong via Wikimedia Commons (CC BY-SA 4.0)	https://upload.wikimedia.org/wikipedia/commons/thumb/2/2e/Barnes_in_Accra_(2021).jpg/960px-Barnes_in_Accra_(2021).jpg
man_neutral1	CC BY-SA 4.0	Photo by Ponor via Wikimedia Commons (CC BY-SA 4.0)	https://upload.wikimedia.org/wikipedia/commons/thumb/5/5d/Low-key_photography_-_portrait_of_a_young_man.jpg/1200px-Low-key_photography_-_portrait_of_a_young_man.jpg
man_neutral2	CC BY-SA 4.0	Photo by Harjeet Singh Asees via Wikimedia Commons (CC BY-SA 4.0)	https://upload.wikimedia.org/wikipedia/commons/thumb/b/bf/Portrait_of_old_Sikh_man.jpg/960px-Portrait_of_old_Sikh_man.jpg
man_neutral3	CC BY-SA 4.0	Photo by PetarV79 via Wikimedia Commons (CC BY-SA 4.0)	https://upload.wikimedia.org/wikipedia/commons/thumb/6/63/Petar_Vukasinovic.jpg/960px-Petar_Vukasinovic.jpg
dog_chain_guard1	Pexels License	Photo by Phan Cuong on Pexels.com (free use)	https://images.pexels.com/photos/26509903/pexels-photo-26509903.jpeg
dog_chain_guard2	Pexels License	Photo by Ivan Babydov on Pexels.com (free use)	https://images.pexels.com/photos/7789879/pexels-photo-7789879.jpeg
dog_chain_guard3	Pexels License	Photo by Phan Cuong on Pexels.com (free use)	https://images.pexels.com/photos/26509926/pexels-photo-26509926.jpeg
dog_neutral1	Pexels License	Photo by Tomris on Pexels.com (free use)	https://images.pexels.com/photos/29357138/pexels-photo-29357138.jpeg
dog_neutral2	Pexels License	Photo by Hardial Singh on Pexels.com (free use)	https://images.pexels.com/photos/32098784/pexels-photo-32098784.jpeg
dog_neutral3	Pexels License	Photo by Reelssman on Pexels.com (free use)	https://images.pexels.com/photos/30673846/pexels-photo-30673846.jpeg